

Table-of-Marks Separation of a Primitive Four-versus-Four Burnside Relation

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Abstract

We give an exact, support-restricted table-of-marks certificate for a primitive Burnside relation arising from the $W(E_6)$ lambda-square atlas. The 16 core-free subgroup types $S_1, \dots, S_{16}$ have a 16×16 integral self-mark matrix of rank 16 and determinant $226492416 = 2^{23}3^3$. The relation found by permutation-character linearization,

$$R_4 = X_2 + X_3 + X_5 + X_6 - X_{11} - X_{12} - X_{13} - X_{14},$$

therefore does not vanish in the corresponding Burnside submodule. Its mark vector is explicit and has content two. All values are reconstructed from frozen permutation source bytes and replayed by an independent checker. We make no claim about the full Burnside ring or about arithmetic Euler factors.

1 Question and scope

Let $G = W(E_6)$ be the group of order 51840 supplied by the preceding exact group package. The C62 atlas supplies sixteen ambient-conjugacy classes of core-free subgroups, denoted $S_1, \dots, S_{16}$, and C63 studies the permutation characters of the transitive G -sets $X_i = G/S_i$. Its exterior square shadow gives a rational character relation R_4 as displayed above. The present question is whether this relation is a genuine equality in the Burnside ring, or only after applying the linearization map to characters. The distinction is the classical mark-versus-character distinction [1, 3, 2].

We restrict throughout to the submodule generated by these sixteen types. The literal scope firewall is

NO_BAD_EULER_OR_ROOT_NUMBER.

In particular, no decomposition groups, resolvents, local fields, bad Euler factors, root numbers, automorphy, or Hilbert–Polya operator are inferred.

2 Marks and exact reconstruction

For a subgroup H and a transitive set G/K , the mark is

$$\Phi_H(G/K) = |(G/K)^H| = \frac{\#\{g \in G : g^{-1}Hg \subseteq K\}}{|K|}.$$

The quotient is integral because the numerator is a union of right K -cosets. Define the restricted mark matrix M by

$$M_{ij} = \Phi_{S_i}(G/S_j), \quad 1 \leq i, j \leq 16.$$

The producer first reads the C61 generator arrays and closes them under composition, obtaining exactly 51840 permutations of 27 points. It then reads every C62 atlas stabilizer, verifies its digest, order, and core order, and selects the dictionary representative for each S_i . For every ordered pair (i, j) it tests containment after conjugating a deterministic generating set of S_i by each of the 51840 group elements. This is a direct count, not a lookup of a table-of-marks package.

3 The mark certificate

Theorem 1 (16-type separation). *For the frozen $W(E_6)$ subgroup representatives, the exact matrix M has*

$$\text{rank}_{\mathbb{Q}} M = 16, \quad \det M = 226492416 = 2^{23}3^3.$$

Consequently the mark map is injective on $\mathbb{Q}\{X_1, \dots, X_{16}\}$.

The rows of M , in the order $S_1, \dots, S_{16}$, are

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 \\ 7 & 12 & 12 & 24 & 0 & 0 & 36 & 0 & 0 & 0 & 12 & 12 & 0 & 0 & 14 \\ 6 & 12 & 14 & 2 & 0 & 0 & 0 & 6 & 0 & 0 & 2 & 4 & 0 & 0 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 16 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 8 & 0 & 0 & 0 & 8 & 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

Proposition 1 (The C63 relation is mark-visible). *For $r_4 = (0, 1, 1, 0, 1, 1, 0, 0, 0, 0, -1, -1, -1, -1, 0, 0)^T$,*

$$Mr_4 = (0, 4, 2, 0, 0, -2, 0, 20, 0, 0, -2, -4, 2, 0, 0, 0)^T.$$

Thus R_4 is nonzero in the restricted Burnside module; its second mark is four and the content of the image is two.

The distinction is conceptually useful. C63's character map forgets exactly the three displayed rational directions on this support, while the mark map retains stabilizer-fixed-point information. The determinant gives a compact integral certificate that no additional relation occurs among these sixteen transitive sets.

4 Verification protocol

The canonical evidence file is produced by `code/c64_mark.py`. Its source hash bindings are

source	SHA-256
C61 group evidence	f4be3a2c5990120a97264505ba9f21b55b8f8c330521044936a52f68e8cd89e9
C62 atlas	3c40a674779a0e2d6d1c69b7c4ccc9115c4b9a2ba049684e8c5204b94b621c40
C62 dictionary	85dc05d043e3781631330083303b98995201c7c30b53aae2901d5fd874b6cd5e
C63 evidence	38f439cfe6ed71616a7c74d68bd07da73f5680566ae16f8c557ab2b5d1d16e26

The independent checker does not import the producer. It reconstructs the group and subgroup sets, repeats all 256 containment counts, recomputes the rank and fraction-free determinant, and checks the relation image. A clean process replay returns `REPLAY_PASS`; ten mutations of status, scope, source hashes, matrix entries, scalar invariants, relation signs, and claims are all rejected.

5 Limitations and successor boundary

The full table of marks of $W(E_6)$ contains more subgroup classes than the sixteen types used here. Injectivity of this restricted matrix therefore does not imply a classification of the full Burnside ring. Nor does a mark distinction by itself distinguish arithmetic fields at bad primes or produce an Euler product. Those questions are deliberately left outside the scope of this paper.

6 Conclusion

The C63 four-versus-four equality is a character-level shadow, not a Burnside equality: its exact mark witness is already visible at the S_2 row. The 16-type mark matrix is nonsingular, providing a short, reproducible separation certificate and a controlled starting point for any later integral question.

References

- [1] Andreas W. M. Dress. *Contributions to the theory of induced representations*. Springer, 1991.
- [2] Fritz Gassmann. Bemerkungen zu der vorstehenden arbeit von hurwitz. *Math. Z.*, 25:665–675, 1926.
- [3] Jean-Pierre Serre. *Linear Representations of Finite Groups*. Springer, 1977.